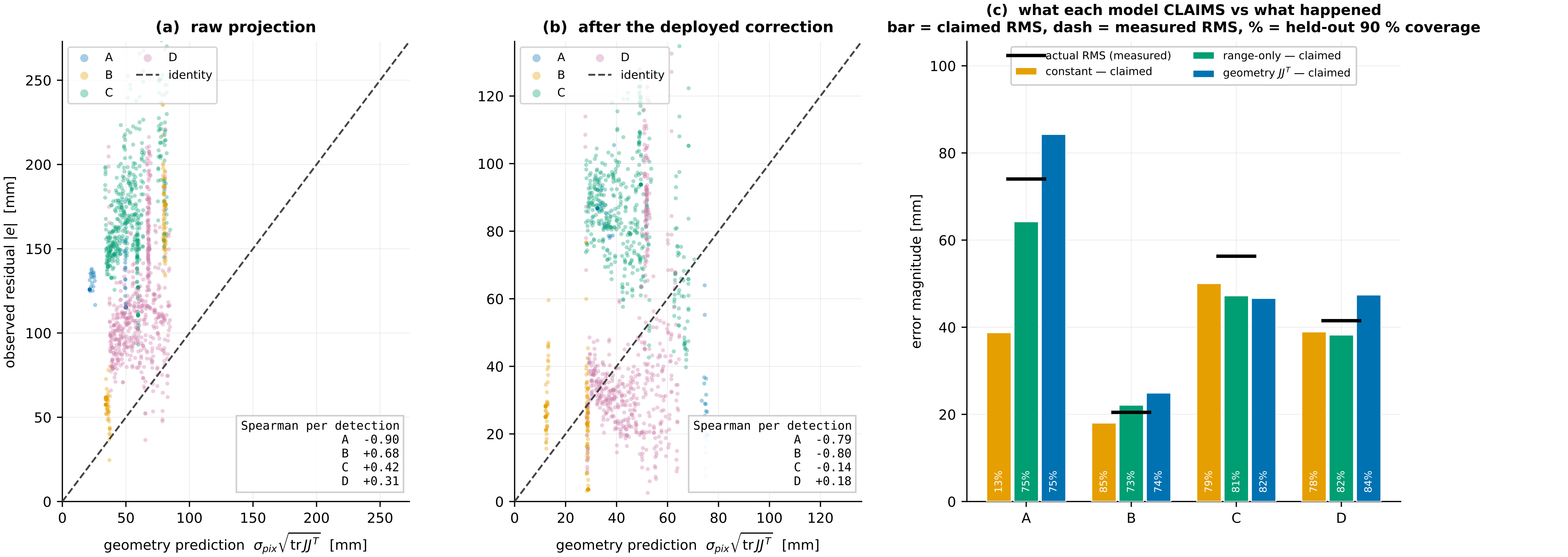
Does projection geometry explain the observed residual? One point per detection, no range binning



1424 detections from THREE captures (smoke1 90 deg, smoke2 0 deg, fusion_handover tangent-derived): two headings only. The robot drove straight routes, so the sampled footprint is a thin ribbon -- image column spans ~100 px of ~1280 on cameras A and B -- and sigma_max(J) is collinear with range at rho = 0.976/0.969/0.999/0.996. This data therefore CANNOT test whether JJ^T adds anything beyond distance from camera; the range-only bar is shown so that limit is visible rather than assumed. Folds are leave-region-out, not train/val/test by run. Ground truth measures the residual and never enters a projection, Jacobian or covariance.